John Gargalionis

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Citizenship: Australian • Inspire: J.Gargalionis.1

Experience

Nov 2024 – Current | **Postdoctoral Researcher (ARC Centre of Excellence for Dark Matter Particle Physics, Adelaide University)**
- Contact officer for Adelaide node of dark-matter centre

Jan 2022 – Nov 2024 | **Investigador Doctor Senior (IFIC and the University of Valencia, Spain)**
- Awarded the Juan de la Cierva Fellowship
- Organiser of Machine Learning Journal Club

May 2021 – Oct 2021 | **Research Associate (The University of Melbourne)**
- Taken in between the end of PhD and the beginning of first postdoctoral position

Education

2016 – 2020 | **The University of Melbourne**
PhD in Theoretical Particle Physics
Advisor: Prof. Raymond Volkas
Thesis: Models of radiative neutrino mass and lepton flavour non-universality
- Doctoral work awarded Chancellor's Prize for Excellence in the PhD

2014 – 2016 | **The University of Melbourne**
MSc in Theoretical Particle Physics (First Class Honour)
Advisors: Prof. Raymond Volkas & Prof. Elisabetta Barberio
Thesis: Neutrino mass through leptoquarks: a new radiative model and its experimental prospects
- Awarded the Prof. Kernot Scholarship in Physics (highest mark in cohort)
- Awarded the N. D. Goldsworthy Scholarship for Physics (excellence in coursework)
- Received the Science Abroad Travel Scholarship (funded a research visit to Europe)
- MSc chosen over fully supported place in Melbourne MD (graduate medicine)

2010 – 2014 | **The University of Melbourne**
BSc with a Physics major (First Class Honour)
- Additional specialisations in Classics (Ancient Greek and Latin) and Neuroscience

Publications

Mar 2026 **Analytic formulae for non-local magic in bipartite systems of qutrits and ququints**
Giorgio Busoni, JG, Ewan N. V. Wallace, Martin J. White
arXiv:2603.09155, Preprint

Mar 2026 **Non-local nonstabiliserness in Gluon and Graviton Scattering**
JG, Nathan Moynihan, Michael L. Reichenberg Ashby, Ewan N. V. Wallace, Chris D. White, Martin J. White
arXiv:2603.04148, Preprint

Aug 2025 **Spin versus nonstabilizerness in gluon and graviton scattering**
JG, Nathan Moynihan, Sokratis Trifinopoulos, Ewan N. V. Wallace, Chris D. White, Martin J. White
arXiv:2508.14967, PhysRevD.113.016007

Jul 2025 **Probing Double-Peaked Gamma-Ray Spectra from Primordial Black Holes with Next-Generation Gamma-Ray Experiments**
C. J. Ouseph, Giorgio Busoni, JG, Sin Kyu Kang, Anthony G. Williams
arXiv:2507.16244, Submitted to JHEP

Jun 2025 **Emergent symmetry in a two-Higgs-doublet model from quantum information and nonstabilizerness**
Giorgio Busoni, JG, Ewan N. V. Wallace, Martin J. White
arXiv:2506.01314, PhysRevD.112.035022

Mar 2025 **Squeezing proton decay and neutrino masses: upper bounds on standard model extensions**
Arnau Bas i Beneito, JG, Juan Herrero–García & Michael A. Schmidt
arXiv:2503.20928, JHEP10(2025)083

Dec 2024 **Linear Standard Model extensions in the SMEFT at one loop and Tera-Z**
JG, Jérémie Quevillon, Pham Ngoc Hoa Vuong & Tevong You
arXiv:2412.01759, JHEP07(2025)136

Jan 2024 **Model-independent estimates for loop-induced baryon-number-violating nucleon decays**
JG, Juan Herrero–García & Michael A. Schmidt
arXiv:2401.04768, JHEP06(2024)182

Dec 2023 **An EFT approach to baryon number violation: lower limits on the new physics scale and correlations between nucleon decay modes**
Arnau Bas i Beneito, JG, Juan Herrero–García, Arcadi Santamaría & Michael A. Schmidt
arXiv:2312.13361, JHEP07(2024)004, JHEP02(2026)065 (erratum)

Dec 2023 **Quark-versus-gluon tagging in CMS Open Data with CWoLa and TopicFlow**
Matthew J. Dolan, JG & Ayodele Ore
arXiv:2312.03434, JHEP08(2025)024

Nov 2022 **Dimension-5 baryon-number violation in low-scale Pati–Salam**
Tomasz P. Dutka & JG
arXiv:2211.02054, PhysRevD.107.035019

Jan 2021 **Exploding operators for Majorana neutrino masses**
JG & Raymond R. Volkas
arXiv:2009.13537, JHEP01(2021)074

May 2020 **Solutions to Problems at Les Houches Summer School on EFT**
Marcel Balsiger, Marios Bounakis, Mehdi Drissi, JG, Erik Gustafson, Greg Jackson, Matthew Leak, Christopher Lepenik, Scott Melville, Daniel Moreno, Michele Tammaro, Selim Touati & Timothy Trott
arXiv:2005.08573, published as an appendix to the lecture notes

Dec 2019 **Radiative neutrino mass model from a mass dimension 11 $\Delta L = 2$ effective operator**
JG, Iulia Popa-Matteiu & Raymond R. Volkas
arXiv:1912.12386, JHEP03(2020)150

Jun 2019 **A near-minimal leptoquark model for reconciling flavour anomalies and generating radiative neutrino mass**
Innes Bigaran, JG & Raymond R. Volkas
arXiv:1906.01870, JHEP10(2019)106

Apr 2017 **Reconsidering the one leptoquark solution: flavour anomalies and neutrino mass**
Yi Cai, JG, Michael A. Schmidt & Raymond R. Volkas
arXiv:1704.05849, JHEP10(2017)047

Apr 2016 **Explaining the 750 GeV diphoton excess with a coloured scalar charged under a new confining gauge interaction**
Robert Foot & JG
arXiv:1604.06180, PhysRevD.94.011703

Talks and seminars

Apr 2026 Quantum Observables for Collider Physics 2026, CERN
Invited plenary talk; unable to attend for personal reasons. Giorgio Busoni presented in my place using my slides, Slides

Feb 2026 ARC Centre of Excellence for Dark Matter Particle Physics Theory Annual Workshop, Canberra
Invited talk

Dec 2025	International Joint Workshop on the Standard Model and Beyond 2025, Taipei *Invited plenary talk*
Dec 2024	Standard Model and Beyond 2024 & 3rd Gordon Godfrey Workshop on Astroparticle Physics *Invited plenary talk*, Slides
Oct 2024	International Workshop on Baryon and Lepton Number Violation (BLV) 2024, Karlsruhe *Contributed plenary talk*, Slides
Sep 2024	Higgs Hunting 2024, Orsay and Paris *Invited plenary talk*, Slides
Jun 2024	SUSY 2024, Madrid *Contributed parallel talk*, Slides
Jun 2024	Planck 2024, Lisbon *Contributed parallel talk*, Slides
Feb 2024	Laboratoire de Physique de l'Ecole Normale Supérieure (LPENS), Paris *Invited seminar*
Aug 2023	The University of Melbourne *Invited seminar*, Details
Jul 2023	University of Tokyo *Invited online seminar*
Jun 2023	Higgs and Effective Field Theory (HEFT) 2023, Manchester *Contributed talk*, Slides
Jun 2023	University of Basel *Invited seminar*
May 2023	Planck 2023, Warsaw *Contributed parallel talk*, Slides
Oct 2022	Korean Institute for Advanced Study (KIAS), Seoul *Invited seminar*, Recording on YouTube
Jun 2022	Higgs and Effective Field Theory (HEFT) 2022, Granada *Contributed talk*, Slides
May 2022	Instituto de Física Corpuscular (IFIC), Valencia *Contributed seminar*
Nov 2020	University of Melbourne *PhD completion seminar*
Oct 2018	Belle II theory interface platform, KEK, Japan *Invited lecture*

May 2018	Monash University, Melbourne *Invited seminar*
Aug 2017	Technische Universität Dortmund *Contributed seminar*
Aug 2017	Technische Universität München *Contributed seminar*
May 2017	Instant workshop on *B*-meson anomalies, CERN *Invited talk*, Recording on CDS
Dec 2016	APPC-AIP Congress, Brisbane *Contributed parallel talk*

Research visits

Dec 2025 IFIC and University of Valencia
Month-long research visit

Aug 2023 University of Melbourne
Month-long research visit

Aug 2022 University of Melbourne
Month-long research visit

May 2022 Laboratory of Subatomic Physics & Cosmology (LPSC), Grenoble
Two-week invited research stay

Honours, scholarships and fellowships

2022 Chancellor's Prize for Excellence in the PhD (The University of Melbourne)
Awarded each year to up to only seven nominees for outstanding doctoral work.

2022 Juan de la Cierva Fellowship (Ministerio de Ciencia e Innovación, Spain)
Two years' funding for my own research programme.

2018 Science Abroad Travel Scholarship (University of Melbourne)

2016 Australian Postgraduate Award (Australian Federal Government)

2015 Prof. Kernot Research Scholarship in Physics (The University of Melbourne)
Awarded to the graduate from the MSc in Physics with the highest grade in research and coursework for that year that continued on to doctoral studies.

2014 N. D. Goldsworthy Scholarship (The University of Melbourne)
Awarded for excellence in graduate coursework.

Supervision and teaching

2025 **Honours co-supervision: Scott Smith**
Co-supervised student on project in EFT, baryon- and lepton-number violation.

Sep 2023 **Invited discussion leader: European School of High Energy Physics (Denmark)**
Invited again as discussion leader at CERN school. Could not attend for personal reasons.

2023 – 2024 **PhD co-supervision: Arnau Bas i Beneito**
Co-supervised student on topics in EFT, baryon- and lepton-number violation and model building.

Oct 2022 **Invited discussion leader: Asia-Europe-Pacific School of High Energy Physics (AEPSHEP) (South Korea)**
Ran daily discussion sessions with graduate students reviewing lecture materials.

2022 **MSc co-supervision: Elena Bermejo Martínez**
Thesis on baryon- and lepton-number violation in simple BSMEFTs. Has left the field and found employment in the energy sector.

Sem 1, 2020 & 2021 **Teaching assistant, PHYC90008: Quantum Field Theory (The University of Melbourne)**
Tasks included writing assignments and solutions.

Sem 1, 2020 **High-school teacher (Victorian School of Languages)**
Taught Modern Greek to students in their last year of secondary school, transitioned students to online learning during initial stages of the pandemic.

Jan 2018 **Tutor: Advanced Scientific Programming in Python Asia–Pacific School (Melbourne)**
Included topics: version control using Git; Python parallelism; context managers and generators; data cleaning and visualisation; Cython, numba and code profiling.

2014 – 2019 **Tutor (The University of Melbourne)**
Tutor and grader for a number of undergraduate subjects taught by the school of physics including: Quantum Physics (PHYC30018), Computational Physics (PHYC20013 and PHYC30021), Subatomic Physics (PHYC30011), for which I also redesigned the tutorial worksheets, and first-year physics (PHYC10003). I have also demonstrated labs for first-year and third-year (PHYC30021) physics.

2014 – 2016 **Curriculum designer and lecturer (Centre for Adult Education)**
Modern and Ancient Greek

Scientific Outreach

Aug 2025 Outreach training (Australian Space Discovery Centre)
Introduced dark-matter outreach programme to educators outside of the research centre.

2017 Physics Workshops (Hume Central Secondary School)
Designed and coordinated a series of five workshops with secondary school students.

2016 Undergraduate seminar series (University of Melbourne)
Presented research work to undergraduate physics students.

2015, 2018 & 2019 CoEPP Work Experience Programme
Presented introductory talk on the SM, sat on organising committee.

2015 & 2016 International Masterclass in Particle Physics

Skills

Programming languages
- Proficient in: Python and Mathematica
- Familiar with: Haskell, C/C++, Clojure and Scheme

Libraries and frameworks
- Scientific Python: NumPy, SciPy, SymPy, Pandas, Jupyter, *etc.*
- High-quality data visualisation with Matplotlib and Seaborn
- Machine learning with PyTorch and TensorFlow
- Version control using Git
- Experienced in cluster computing and job submission with PBS and Slurm

Pheno
- Confident in analysing the viability of BSM models

Languages English (native), Greek (fluent), Spanish (intermediate)